JSS 2 MATHEMATICS LESSON NOTES (THIRD TERM)

WEEK 2: ANGLES AND POLYGONS

TOPIC:

Definition of Angles; Construction of Angles; Polygons and Interior Angles.

A. ANGLES – DEFINITION AND TYPES

Definition

An **angle** is the amount of turn between two straight lines that meet at a point called the **vertex**.

Parts of an Angle

- **Vertex:** Point where the lines meet
- **Arms (Rays):** The two lines forming the angle
- **Notation:** $\angle AOB$ (where O is the vertex)

Measuring Angles

- Unit: **Degrees (°)**
- Tool: **Protractor**
- Full turn = **360°**
- Straight line = **180°**
- Right angle = **90°**

B. TYPES OF ANGLES

1. **Acute Angle**
   - $0^{\circ} < angle < 90^{\circ}$
   - Examples: 30°, 45°, 60°, 75°
2. **Right Angle**
   - Exactly **90°**
3. **Obtuse Angle**
   - $90^{\circ} < angle < 180^{\circ}$
   - Examples: 105°, 120°, 135°, 150°
4. **Straight Angle**
   - Exactly **180°**
5. **Reflex Angle**
   - $180^{\circ} < angle < 360^{\circ}$
   - Examples: 210°, 270°, 315°
6. **Complete (Full) Angle**
   - Exactly **360°**

C. ANGLE RELATIONSHIPS

1. Complementary Angles

- Sum = **90°**
- Example: 30° + 60° = 90°
- Complement of 35° = **55°**

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2. Supplementary Angles

- Sum = **180°**
- Example: 110° + 70° = 180°
- Supplement of 125° = **55°**

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3. Vertically Opposite Angles

- Formed when two lines intersect
- Opposite angles are **equal**

Properties:

- $\angle A = \angle C$
- $\angle B = \angle D$

- Adjacent angles = **180°**

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4. Adjacent Angles

- Share a **common vertex and arm**

D. CONSTRUCTING ANGLES

Using a Protractor (Direct Method)

Example: Construct **75°**

1. Draw baseline AB
2. Place protractor at A
3. Mark 75°
4. Join mark to A
5. Label angle

Using Compass and Ruler

Basic Constructible Angles

- 60°, 90°, 120°, 30°, 45°

I. <u>Constructing 60°:</u>

Steps: 1. Draw line AB With A as center, draw arc

3. With same radius from arc intersection, draw another arc

4. Join A to intersection point Result: ∠ = 60°

<u>II . Constructing 90° (Perpendicular):</u>

Steps: 1. Draw line AB

2. From point A, draw arcs above and below line

3. From both intersections, draw arcs to intersect

4. Join A to intersection point Result: ∠ = 90° Constructing 30°: Construct 60°, then bisect it 3.

<u>III . Constructing 30°</u>

Bisect 60°

<u>IV. Constructing 45°</u>

- Bisect 90°

V. Other Angles

- 75° = 60° + 15°
- 105° = 60° + 45°
- 120° = 60° + 60°
- 135° = 90° + 45°

E. POLYGONS

Definition

A **polygon** is a closed plane figure made up of **three or more straight sides**.

Parts

⬣ Sides: The straight line segments ⬣ Vertices: Points where sides meet ⬣ Interior angles: Angles inside the polygon ⬣ Exterior angles: Angles outside the polygon ⬣ Diagonals: Lines joining non-adjacent vertices

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Types by Number of Sides

Name	Sides
Triangle	3
Quadrilateral	4
Pentagon	5
Hexagon	6

Name	Sides
Heptagon	7
Octagon	8
Nonagon	9
Decagon	10
Dodecagon	12

Types of Polygons by Regularity and shape

1. **Regular Polygon**

All sides equal   All angles equal   Examples: Equilateral triangle, square, regular pentagon

Irregular Polygon

Sides and/or angles not all equal   Examples: Rectangle, scalene triangle

Convex Polygon

All interior angles < 180°   All vertices point outward

Concave Polygon

- At least one interior angle > 180°

F. SUM OF INTERIOR ANGLES

Formula:

Sum of interior angles = (n - 2) × 180° OR Sum = (2n - 4) right angles (since 1right angle = 90°)Where **n = number of sides**

Formula: Sum of interior angles = (n - 2) × 180° OR Sum = (2n - 4) right angles (since 1right angle = 90°)

Where n = number of sides

Derivation: A polygon with n sides can be divided into (n - 2) triangles from one vertex. Each triangle has angle sum = 180° Therefore, total = (n - 2) × 180° Example 6: Triangle (n = 3)

Sum = (3 - 2) × 180° = 1 × 180° = 180°

Example 7: Quadrilateral (n = 4)

Sum = (4 - 2) × 180° = 2 × 180° = 360°

Example 8: Pentagon (n = 5)

341

Sum = (5 - 2) × 180° = 3 × 180° = 540°

Example 9: Hexagon (n = 6)

Sum = (6 - 2) × 180° = 4 × 180° = 720°

Example 10: Octagon (n = 8)

Sum = $(8 - 2) \times 180° = 6 \times 180° = 1{,}080°$

G. EACH INTERIOR ANGLE OF REGULAR POLYGON

Examples:

Example 1: Each angle in regular pentagon
$n = 5$
Each angle = $[(5 - 2) \times 180°] / 5$
$= [3 \times 180°] / 5$
$= 540° / 5 = 108°$
$n = 5$ Each exterior angle = $360° / 5 = 72°$
Check: Interior + Exterior = 180°
i .e. $108° + 72° = 180°$ ✓

Example 2: Each angle in regular hexagon
$n = 6$
Each angle = $[(6 - 2) \times 180°] / 6$
$= 720° / 6 = 120°$

Example 3: Each angle in regular octagon
$n = 8$
Each angle = $[(8 - 2) \times 180°] / 8 = 1{,}080° / 8$
$= 135°$

H. EXTERIOR ANGLES

Key Facts:

- Interior + Exterior = **180°**
- Sum of all exterior angles = **360°**

Formula (Regular Polygon):

Each exterior angle = **360° / n**= 72 AND interior angles = **[(n-2)180]/n** = 108

Example 14: Each exterior angle of regular pentagon n = 5 Each exterior angle = 360° / 5 = 72°

Check: Interior + Exterior = 180°   108° + 72° = 180° ✓

Example 15: Each exterior angle of regular hexagon = 360° / 6 = 60° AND interior angles = **[(n-2)180]/n** = 120 i.e**[(6-2)180]/6 =120**

Check: Interior + Exterior = 180° i.e   60° + 120° = 180° ✓

I. FINDING NUMBER OF SIDES

If given each interior angle:

Example 1: Each interior angle is 140°. Find number of sides.

Method 1: Using formula Each angle = $[(n - 2) \times 180°] / n$

$140 = [(n - 2) \times 180] / n$

$140n = 180n - 360$

$40n = 360$

$n = 9$ The polygon has 9 sides (nonagon)

Example 2: Using exterior angle Each interior angle = 140°

Each exterior angle = $180° - 140° = 40°$

$n = 360° / 40° = 9$ sides .

J. PROBLEM SOLVING

.Example 1: A polygon has 12 sides. Find: a) Sum of interior angles b) Each interior angle if regular c) Each exterior angle if regular
Solution:

Sum = $(12 - 2) \times 180° = 10 \times 180° = 1,800°$

b) Each interior angle = $1,800° / 12 = 150°$

c) Each exterior angle = 360° / 12 = 30° OR: 180° - 150° = 30° 344

Example 2: Four angles of a pentagon are 100°, 110°, 105°, and 95°. Find the fifth angle.

Solution: Sum of interior angles = (5 - 2) × 180° = 540°

Sum of 4 angles = 100° + 110° + 105° + 95° = 410° Fifth angle = 540° - 410° = 130°

Example 3: In a hexagon, three angles are each 130° and two are each 110°. Find the sixth angle.

Solution: Sum = (6 - 2) × 180° = 720° Known angles = 3(130°) + 2(110°) = 390° + 220° = 610° Sixth angle = 720° - 610° = 110°

Example 4: Each exterior angle of a regular polygon is 24°. How many sides does it have?

Solution: n = 360° / exterior angle n = 360° / 24° n = 15 sides

Example 5: Can a regular polygon have each interior angle of 140°? Check: Exterior angle = 180° - 140° = 40° n = 360° / 40° = 9 ✓ Yes, it's a regular nonagon (9 sides) 345

Example 6: Can a regular polygon have each interior angle of 170°? Check: Exterior angle = 180° - 170° = 10° n = 360° / 10° = 36 ✓ Yes, it's a 36-sided polygon

CLASS WORK

Define an angle

What is an acute angle?

Find complement of 37°

Find supplement of 115°

State formula for interior angles

Sum of heptagon angles

Interior = 120°, find sides

Find missing pentagon angle

Exterior = 30°, find sides

Describe construction of 75°

REVISION QUESTIONS

1. List 5 types of angles
2. Difference between complementary & supplementary
3. When are angles vertically opposite?
4. Define polygon
5. List first 10 polygons
6. Regular vs irregular polygon
7. Formula for interior angle
8. Why exterior sum = 360°?
9. Construct 60°
10. Can polygon have 100° interior angle?

ASSIGNMENT

Section A: Angle Types

- Classify angles
- Find complements and supplements
- Solve angle ratios

Section B: Construction

Construct:

- 60°, 90°, 45°, 30°, 75°, 105°, 120°, 135°

Section C: Polygons Table

Complete:

- Triangle → Decagon (angles & sums)

Section D: Problem Solving

- Missing angles
- Regular polygons
- Ratio problems
- Interior/exterior calculations

Section E: Mixed Problems

- Quadrilateral angles
- Polygon identification
- Triangle problems
- Proof: Exterior sum = 360°